

Customer Shopping Behavior Analysis

1. Project Overview

Customer Trends Analysis is a data analytics project that explores customer purchasing behavior using Python, Pandas, NumPy, Matplotlib, and Seaborn. The project performs data cleaning and exploratory data analysis (EDA) to identify patterns in customer demographics, spending habits, and sales trends. Various visualizations are used to uncover business insights and customer preferences. The analysis helps understand factors affecting revenue and customer behavior, enabling data-driven decisions for marketing, customer retention, and overall business growth. This project demonstrates practical skills in data preprocessing, visualization, and business analytics.

Dataset Summary

1. Dataset Size

- Contains 3,900 customer records.
- Includes 18 features related to customer demographics and shopping behavior.

2. Customer Information

- Contains demographic attributes such as Age, Gender, and Location.
- Helps analyze customer distribution across different regions and groups.

3. Shopping Details

- Includes Item Purchased, Category, Purchase Amount (USD), Season, and Frequency of Purchases.
- Provides insights into purchasing habits and product preferences.

4. Transaction Information

- Records Payment Method, Discount Applied, and Subscription Status.
- Enables analysis of payment preferences and promotional impact.

5. Customer Feedback

- Includes Review Rating to evaluate customer satisfaction.
- Useful for understanding customer experience and purchasing behavior.

3. Exploratory Data Analysis (EDA)

1. Data Cleaning

- Checked for missing values and duplicate records.
- Verified data types to ensure data consistency and quality.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard	Yes	Yes	14	Venmo	Weekly
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes	Yes	49	Cash	Quarterly
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping	Yes	Yes	19	Credit Card	Weekly
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes	Yes	8	Venmo	Annually
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping	Yes	Yes	4	Cash	Quarterly

2. Statistical Analysis

- Generated descriptive statistics for numerical features.
- Analyzed Age, Purchase Amount (USD), and Review Rating to understand customer behavior.

3. Categorical Data Analysis

- Examined Gender, Category, Season, Payment Method, and Location.
- Identified customer distribution and shopping preferences.

4. Data Visualization

- Created bar charts, histograms, pie charts, box plots, and count plots.
- Visualized purchasing trends, product popularity, seasonal demand, spending patterns, and discount usage.

5. Correlation Analysis

- Explored relationships between numerical variables.
- Identified patterns and potential factors influencing customer purchases.

	customer_id	age	purchase_amount_usd	review_rating	previous_purchases	purchase_frequency_days
customer_id	1.000000	-0.004079	0.011048	-0.006718	-0.039159	0.005380
age	-0.004079	1.000000	-0.010424	0.016826	0.040445	0.016544
purchase_amount_usd	0.011048	-0.010424	1.000000	-0.009875	0.008063	0.009526
review_rating	-0.006718	0.016826	-0.009875	1.000000	0.009451	-0.003770
previous_purchases	-0.039159	0.040445	0.008063	0.009451	1.000000	-0.016996
purchase_frequency_days	0.005380	0.016544	0.009526	-0.003770	-0.016996	1.000000

Overall, the EDA transformed raw customer data into actionable insights that can support better marketing strategies, customer segmentation, inventory management, and business decision-making.

- Analyzed revenue by product category, season, and location.
- Identified top-performing categories and spending trends.

5. Payment & Discount Analysis

- Evaluated customer payment preferences.
- Measured the impact of discounts on purchase amounts.

6. Advanced SQL Operations

- Applied GROUP BY, ORDER BY, HAVING, CASE, CTEs, and Window Functions.
- Generated ranked reports and extracted meaningful business insights from the dataset.

Query Query History

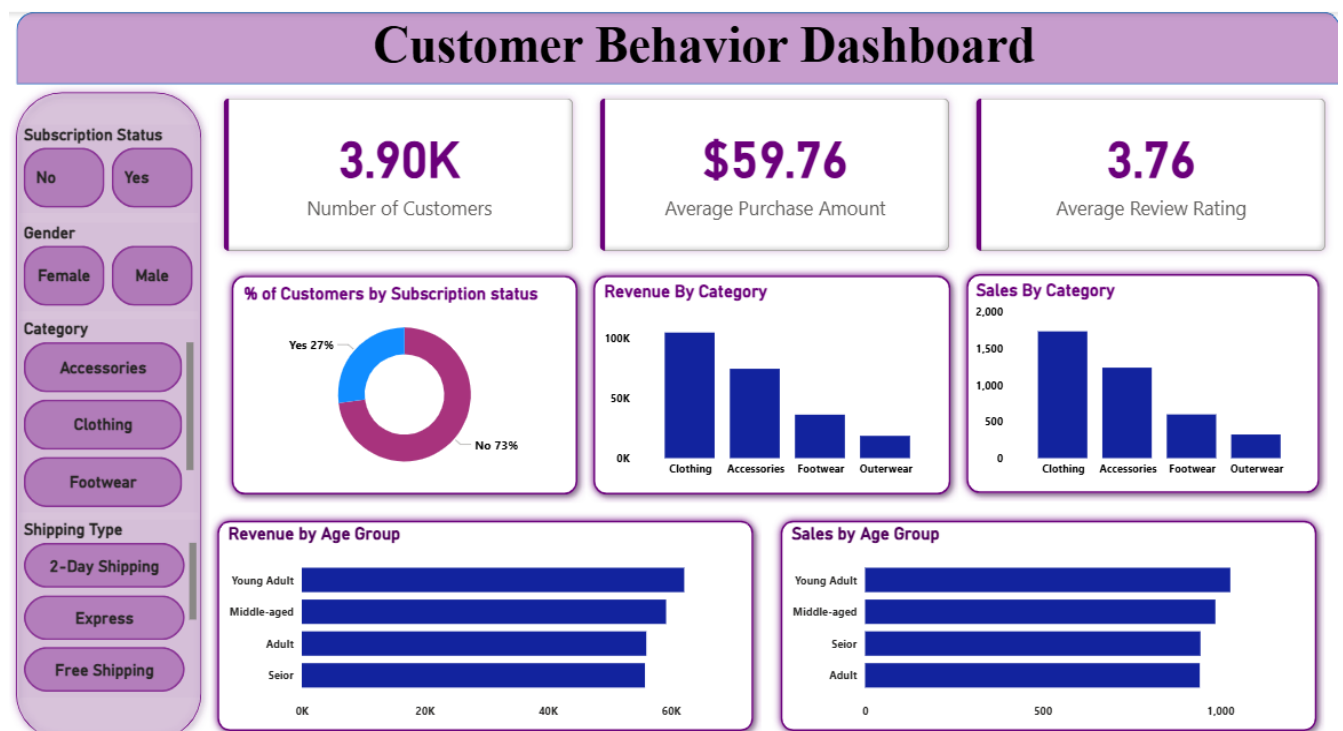
```
44 --7. Segment customers into New, Returnig and Loyal based on their
45 --total number of previous purchases and show the out of each segment.
46 with customer_type as (
47 select customer_id, previous_purchases,
48 CASE
49 WHEN previous_purchases=1 then 'New'
50 WHEN previous_purchases BETWEEN 2 AND 10 then 'Returnig'
51 else 'Loyal' END AS customer_segment
52 from customer
53 )
54 select customer_segment,
55 count(*) as "Number of Customers"
56 from customer_type
57 group by customer_segment;
58
59 --8. what are the top 3 most purchased products within each category?
60 with item_counts as (
61 select category,item_purchased,
62 COUNT(customer_id) as total_orders,
63 ROW_NUMBER() over(partition by category order by count(customer_id)desc) as item_rank
64 from customer
```

Data Output Messages Notifications

customer_segment	Number of Customers
Loyal	3116
New	83
Returnig	701

Total rows: 3 Query complete 00:00:00.122

5. Power BI Dashboard



The Power BI dashboard presents an interactive overview of customer shopping behavior through dynamic charts, KPIs, and filters. It visualizes key metrics such as total sales, customer distribution, product categories, seasonal trends, purchase amounts, and payment methods in a single view. Interactive slicers allow users to filter data by gender, category, location, and season, making it easy to explore trends and compare customer segments. The dashboard transforms raw data into clear, actionable insights that support data-driven business decisions and improve overall analytical efficiency.

6. Business Recommendations

1. Target High-Value Customers

- Identify customers with high purchase amounts and offer personalized rewards, loyalty programs, and exclusive discounts to increase retention.

2. Optimize Inventory

- Stock popular product categories based on customer demand and seasonal purchasing trends to reduce shortages and excess inventory.

3. Improve Marketing Campaigns

- Design targeted marketing campaigns based on customer demographics, location, and shopping preferences to increase sales.

4. Leverage Seasonal Trends

- Launch seasonal promotions and special offers during peak shopping periods to maximize revenue.

5. Enhance Customer Experience

- Monitor review ratings and customer feedback to improve product quality and overall customer satisfaction.

6. Personalize Product Recommendations

- Recommend products based on customers' previous purchases and preferred categories to encourage repeat purchases.

7. Optimize Discount Strategies

- Analyze the impact of discounts on sales and apply promotional offers where they generate the highest return on investment.

8. Expand Preferred Payment Options

- Promote and optimize the most frequently used payment methods to improve customer convenience and reduce transaction friction.

9. Focus on Regional Performance

- Analyze sales across different locations and implement region-specific marketing strategies to improve performance in underperforming areas.

10. Build Customer Loyalty

- Introduce membership programs, reward points, and personalized offers to encourage repeat purchases and strengthen long-term customer relationships.